

Strategic Obfuscation against Adaptive Censorship: Equilibrium Blocking and Dynamic Implementability

Xinhao Dong and Yuqi Hu

Department of Economics, Simon Fraser University

Abstract. Censorship-resistance systems are often evaluated by classifier accuracy and implementation overhead, but deployment outcomes also depend on how an adaptive censor responds. We model a censorship-resistance operator that chooses costly obfuscation and a censor that chooses blocking intensity. Obfuscation changes both effective detection and the benign traffic exposed to collateral blocking, making the censor’s net return to blocking endogenous. We characterize the static equilibrium and then study which obfuscation–blocking targets can be sustained by credible Nash reversion. The relevant incentive constraints use each player’s exact unilateral best response: a prespecified stress test such as disabling obfuscation can miss a profitable partial reduction in effort. In a transparent normalized benchmark, the jointly strictly individually-rational region occupies 6.46% of the target action square; within that region the median and 90th-percentile patience requirements are 0.573 and 0.902 under a uniform target-space measure. The prespecified CIR deviation changes the binding combined constraint on 18.62% of that region and can reverse sustainability classifications. A public-signal extension shows how missed detections and false alarms further tighten implementability. The analysis separates making traffic difficult to classify from making an obfuscation regime strategically stable against an adaptive censor.

Keywords: Censorship resistance · Game theory · Traffic obfuscation · Adaptive censorship · Repeated games · Collateral damage

1 Introduction

A censorship-resistance system (CRS) can make its traffic harder to classify without making it safe to deploy. A censor observes traffic, decides how aggressively to block it, and trades successful suppression against collateral damage to benign users and services. The CRS operator in turn chooses how much costly obfuscation to maintain. Traffic transformation therefore changes a strategic payoff, not only a classifier score: it affects both the traffic the censor can identify and the benign traffic implicated by blocking.

This paper models that interaction between a CRS operator (CIR) and an adaptive censor (CEN). The CIR chooses obfuscation intensity $u \in [0, 1]$; the CEN chooses blocking intensity $a \in [0, 1]$. Obfuscation lowers effective detectability $\pi(u)$ and can change benign exposure $T_b(u)$, at cost $K(u)$. The censor’s benefit

from blocking and collateral cost reduce to the net marginal return

$$D(u) = BT_p\pi(u) - \Lambda T_b(u). \quad (1)$$

This statistic is the bridge from system design to censor behavior. A design can discourage blocking by reducing effective detection, by implicating benign traffic that the censor is reluctant to disrupt, or by combining both channels. Blocking simultaneously raises the operator’s return to obfuscation.

The one-shot game establishes which responses are technologically and strategically compatible at a deployment epoch. We characterize pure equilibrium under continuous, possibly nondifferentiable primitives, including the boundary cases in which zero or full blocking is optimal. With strictly convex blocking and obfuscation costs, the opposing response monotonicities yield a unique equilibrium. A transparent exponential-quadratic benchmark has a unique stage equilibrium at $(u^N, a^N) \simeq (0.326, 0.501)$.

The central question is dynamic: when can a desired obfuscation–blocking target remain implementable as both sides repeatedly update their configurations? We use permanent reversion to the selected stage Nash equilibrium as a credible punishment benchmark. For each player, implementability depends on the current best-deviation gain G_i and the continuation surplus L_i destroyed by deviation. In the regular case the patience requirement is $\delta_i = G_i/(G_i + L_i)$. Thus technological feasibility alone is insufficient. Even a small current temptation can require high patience when the target lies only slightly above the credible continuation payoff.

Exact deviation choice matters because obfuscation is continuous and costly. Disabling obfuscation is feasible, but a deviating operator may prefer a partial reduction that retains delivery benefits while saving some cost. In the benchmark, the joint strict-individual-rationality set occupies 6.46% of the normalized target square under a uniform geometric measure. Its median exact patience threshold is 0.573 and its 90th percentile is 0.902. Replacing the exact CIR deviation with the prespecified action $u = 0$ changes the combined threshold on 18.62% of that set and creates false-sustainability classifications for 9.82% of it at $\delta = 0.5$. These shares summarize model geometry, not empirical frequencies.

The paper makes three contributions. First, it makes obfuscation a strategic design choice that changes both effective detectability and benign exposure. This reduces the censor’s response to $D(u)$ and yields a complete characterization of interior and boundary equilibria. Second, it characterizes which targets can be supported by credible Nash reversion. The result separates current temptation from continuation surplus and shows why technical feasibility does not imply dynamic implementability. Third, it solves the operator’s unilateral deviation over the actual continuous action space, establishes when disabling obfuscation is exact, and quantifies the classification error from imposing that action. A public-signal extension shows how false alarms and missed detections tighten these conditions.

The closest censorship-game framework already models strategic blocking, collateral damage, and credible repeated-game punishments [8]. Strategic classification also allows costly manipulation in response to a detector [4,3,11,5]. The

margin advanced here is the endogenous costly obfuscation choice and its target-specific implementability consequences in a censorship game with continuous blocking. Section 2 connects this abstraction to CRS deployment. Sections 3–4 derive the static and dynamic results; Section 5 maps their benchmark geometry; and Section 6 studies monitoring, action granularity, and design implications.

2 System Setting and Related Work

Deployment and Threat Model. A period is a deployment or update epoch long enough for a CRS operator to change its traffic-transformation configuration and for a censor to update blocking aggressiveness. The theory does not assign an epoch a fixed wall-clock duration. The CIR represents the operator, designer, or configurator of a CRS rather than an individual user. Its continuous action u is an aggregate intensity or mixture over available transformation modes. The CEN action a is the aggregate aggressiveness or share of suspicious traffic blocked. It is a reduced-form control, not a claim that every censor exposes a literal continuous knob.

The function $T_b(u)$ is the effective benign traffic exposure associated with blocking transformed traffic. It may represent popular cover services, co-located benign users, or other collateral traffic implicated by a CRS design; the CIR need not create that traffic. Perfect public monitoring means that a sufficiently informative public statistic reveals whether the prescribed action profile was followed. It does not require direct observation of engineering effort. Section 6 replaces this benchmark with an imperfect public signal.

Strategic Censorship and Detection. Elahi et al. [8] model a censor–circumventor interaction with traffic volume, classifier errors, collateral blocking costs, and credible repeated-game continuation. Their circumvention choice changes traffic quantity under a largely given detection technology. Our u instead changes the effective detection and collateral-exposure functions faced by a censor choosing continuous blocking. Other censorship games study decoy routing, proxy assignment, or moving-target infrastructure rather than target support over an obfuscation–blocking action space [17,16,14]. The model-internal deviation result here does not claim that these papers optimize the wrong action.

Adversarial-classification models establish that a classified party can alter features or distributions at a cost in response to a classifier [4,3,11,5,12,15]. Our censorship setting adds a blocking action and an explicit collateral channel. The repeated analysis then asks whether a particular pair of design and blocking choices can be supported by a credible continuation, rather than whether manipulation defeats a fixed or strategically chosen classifier.

Mapping Systems to Primitives. Format transformation and protocol mimicry, as in FTE and Marionette, primarily lower effective detectability $\pi(u)$ at engineering cost $K(u)$ [6,7]. DeltaShaper maps traffic morphing to the same margin [1]. Collateral-leveraging designs such as domain fronting also alter the benign exposure associated with blocking and therefore the $T_b(u)$ channel [9]. Snowflake’s

temporary proxies illustrate adaptation across deployment epochs [2]. Measurement studies further show that effective detection depends on base rates, false positives, and the censor's aggregation unit, so $\pi(u)$ should not be read as a laboratory score independent of deployment context [13,18,19]. The model abstracts from protocol details to isolate how these margins alter an adaptive censor's incentive.

3 Baseline Game and Static Equilibrium

Let $T_p > 0$ denote circumvention demand, $B > 0$ the censor's benefit per blocked unit, $\Lambda > 0$ its marginal collateral cost, and $V > 0$ the CIR value per delivered unit. Write $q \equiv VT_p$. Obfuscation changes effective detection through $\pi(u)$ and benign exposure through $T_b(u)$. Direct blocking and obfuscation costs are $C(a)$ and $K(u)$. The stage payoffs are

$$U_{\text{CEN}}(a, u) = a[BT_p\pi(u) - \Lambda T_b(u)] - C(a), \quad (2)$$

$$U_{\text{CIR}}(a, u) = q[1 - a\pi(u)] - K(u). \quad (3)$$

Define the censor's net return before direct filtering cost as

$$D(u) \equiv BT_p\pi(u) - \Lambda T_b(u). \quad (4)$$

Lower detectability and greater benign exposure both reduce $D(u)$, but they represent distinct engineering channels.

Assumption S (Baseline Regularity). The function $\pi : [0, 1] \rightarrow [0, 1]$ is continuous, nonincreasing, and convex; $T_b : [0, 1] \rightarrow \mathbb{R}_+$ is continuous and nondecreasing; and $C, K : [0, 1] \rightarrow \mathbb{R}$ are continuous and convex, with $C(0) = K(0) = 0$ and K minimized at zero. Convexity of π makes the CIR's own-action problem concave. Differentiability is not assumed; f'_+ and f'_- denote one-sided derivatives of a convex function.

The best-response correspondences can be written

$$\mathcal{A}(d) = \arg \max_{a \in [0, 1]} \{ad - C(a)\}, \quad \mathcal{U}(a) = \arg \max_{u \in [0, 1]} \{q(1 - a\pi(u)) - K(u)\}. \quad (5)$$

Both have nonempty compact interval values. Extremal selections from $\mathcal{A}(D(u))$ are nonincreasing in u , while extremal selections from $\mathcal{U}(a)$ are nondecreasing in a . Strict convexity of C and K makes the two responses single-valued. Appendix A states the complete set-valued KKT conditions.

Proposition 1 (Static equilibrium). *Under Assumption S, a pure Nash equilibrium exists, and (u^*, a^*) is an equilibrium exactly when*

$$u^* \in \mathcal{U}(a^*), \quad a^* \in \mathcal{A}(D(u^*)).$$

The following boundary and interior statements hold.

1. If $D(0) \leq C'_+(0)$, zero blocking is a CEN best response to every u . Hence $(u, 0)$ is an equilibrium for every $u \in \arg \min K$.
2. If $D(1) \geq C'_-(1)$, full blocking is a CEN best response to every u . Hence $(u, 1)$ is an equilibrium for every $u \in \mathcal{U}(1)$, but equilibria with $a < 1$ may also exist.
3. If C and K are strictly convex, the equilibrium is unique. The corresponding strict uniform boundary comparisons in Appendix A determine when zero or full blocking is the unique outcome.
4. If the primitives are differentiable and the equilibrium is interior, it satisfies

$$C'(a^*) = D(u^*), \quad K'(u^*) = qa^*[-\pi'(u^*)]. \quad (6)$$

Boundary equilibria instead satisfy the one-sided conditions in Appendix A.

The weak endpoint comparisons in parts 1 and 2 establish best-response membership, not uniqueness. This distinction matters with flat costs or set-valued responses: $D(1) \geq C'_-(1)$ alone does not imply that every equilibrium has full blocking. In the interior, Eq. (6) balances the censor's net blocking return against marginal filtering cost and the CIR's marginal delivery gain against marginal obfuscation cost.

3.1 Transparent Benchmark

For the numerical illustration, let

$$\pi(u) = \pi_0 e^{-\eta u}, \quad T_b(u) = \tau u, \quad C(a) = \frac{c}{2}a^2, \quad K(u) = \frac{k}{2}u^2. \quad (7)$$

We normalize $B = T_p = V = 1$ and set $\pi_0 = 0.9$, $\eta = 1$, $\Lambda = 0.5$, $\tau = 0.3$, $c = 1.2$, and $k = 1$. These values are illustrative rather than calibrated. The closed-form responses are

$$a^{\text{BR}}(u) = \left[\frac{BT_p\pi_0 e^{-\eta u} - \Lambda\tau u}{c} \right]_{[0,1]}, \quad (8)$$

$$u^{\text{BR}}(a) = \min \left\{ 1, \frac{1}{\eta} W \left(\frac{qa\pi_0\eta^2}{k} \right) \right\}, \quad (9)$$

where W is the principal Lambert- W function. Their unique intersection is $(u^N, a^N) = (0.32555, 0.50091)$, with Nash payoffs $P_{\text{CIR}} = 0.62146$ and $P_{\text{CEN}} = 0.15054$. Independent constrained optimization reproduces the closed forms and the equilibrium to numerical tolerance.

4 Dynamic Implementability

A stage equilibrium describes mutual best responses within one deployment epoch. A CRS may instead seek a target $(\bar{u}, \bar{a})$ that is not a stage equilibrium but could persist because deviation changes future play. We study target-specific support rather than an unrestricted folk-theorem possibility result [10].

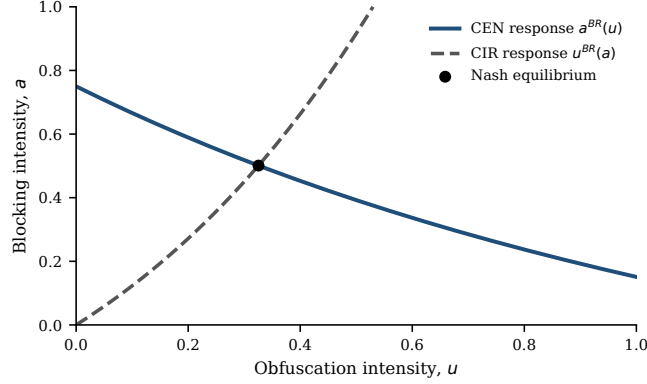


Fig. 1. Best responses in the illustrative benchmark. The CEN response falls as obfuscation lowers $D(u)$; the CIR response rises because blocking raises the delivery benefit from obfuscation. The marker is the unique stage Nash equilibrium.

4.1 Targets, Deviations, and Credible Continuation

The repeated game has common discount factor $\delta \in (0, 1)$ and normalized payoff

$$(1 - \delta) \sum_{t=0}^{\infty} \delta^t U_i(a_t, u_t). \quad (10)$$

Fix a stationary target $(\bar{u}, \bar{a})$. Its stage payoffs are

$$p_{\text{CIR}}^C = q[1 - \bar{a}\pi(\bar{u})] - K(\bar{u}), \quad p_{\text{CEN}}^C = \bar{a}D(\bar{u}) - C(\bar{a}). \quad (11)$$

Each player can choose any unilateral action in the current period. The relevant deviation values are therefore

$$p_{\text{CIR}}^D(\bar{a}) = \max_{u \in [0,1]} U_{\text{CIR}}(\bar{a}, u), \quad p_{\text{CEN}}^D(\bar{u}) = \max_{a \in [0,1]} U_{\text{CEN}}(a, \bar{u}). \quad (12)$$

The target action is feasible in each maximization, so $p_i^D \geq p_i^C$.

The punishment benchmark is permanent reversion to a selected stage Nash equilibrium (u^N, a^N) , with normalized continuation payoff $P_i = U_i(a^N, u^N)$. Repeated Nash play is sequentially rational after every punishment history. By contrast, a minmax payoff does not by itself identify credible stationary punishment actions. Stronger repeated-game punishments may exist, but they are unnecessary for the benchmark result. When the stage game has multiple equilibria, the punishment equilibrium must be specified; the numerical benchmark has a unique one.

Proposition 2 (Support under exact deviations). *Suppose actions are perfectly publicly monitored and a deviation triggers a credible stationary continuation with payoff P_i . The target is supported exactly when, for each player,*

$$p_i^C \geq (1 - \delta)p_i^D + \delta P_i. \quad (13)$$

Because $p_i^D \geq p_i^C$, the cases relevant to exact deviations are:

1. If $p_i^D = p_i^C$ and $P_i \leq p_i^C$, the constraint holds for every δ .
2. If $p_i^D > p_i^C > P_i$, it holds if and only if

$$\delta \geq \delta_i = \frac{p_i^D - p_i^C}{p_i^D - P_i} \in (0, 1). \quad (14)$$

3. If $p_i^D > p_i^C = P_i$, no $\delta < 1$ supports the target.
4. If $P_i > p_i^C$, no $\delta > 0$ supports the target in the continuous-action exact-deviation setting.

The last case includes targets that are current best responses. If $P_i > p_i^C = p_i^D$, actions arbitrarily close to the target lose arbitrarily little today but obtain the strictly better continuation, so every positive δ admits a profitable deviation. Appendix B proves Proposition 2 and gives the complete scalar taxonomy for an arbitrary prespecified current action.

In the regular case, define

$$G_i = p_i^D - p_i^C, \quad L_i = p_i^C - P_i. \quad (15)$$

Then the threshold becomes

$$\delta_i = \frac{G_i}{G_i + L_i}. \quad (16)$$

The deviation gain G_i measures current temptation; the continuation surplus L_i measures what deviation destroys. A high threshold can reflect a large temptation or a target that offers only a thin surplus over Nash reversion. Joint support requires both players' conditions, so $\delta_{\min} = \max\{\delta_{\text{CIR}}, \delta_{\text{CEN}}\}$ whenever both lower thresholds are defined.

4.2 Why the Exact Obfuscation Deviation Matters

For target blocking $\bar{a}$, write the CIR deviation objective as

$$\phi_{\bar{a}}(u) = q[1 - \bar{a}\pi(u)] - K(u). \quad (17)$$

Let $d^{\text{exact}} = \max_{u \in [0,1]} \phi_{\bar{a}}(u)$ and let $d^0 = \phi_{\bar{a}}(0)$ be the value from disabling obfuscation. The latter is a useful stress test, but it need not be the best deviation.

Proposition 3 (Exact versus zero-obfuscation deviations). *Under Assumption S, $d^{\text{exact}} \geq d^0$, with equality exactly when $0 \in \mathcal{U}(\bar{a})$. Disabling obfuscation is optimal if and only if*

$$q\bar{a}[-\pi'_+(0)] \leq K'_+(0). \quad (18)$$

If this inequality is reversed strictly, then $d^{\text{exact}} > d^0$. In particular, with $\bar{a} > 0$, $\pi'_+(0) < 0$, and $K'_+(0) = 0$, zero obfuscation is not a best response.

The condition compares the marginal cost saved at zero with the marginal delivery benefit from retaining a small amount of obfuscation. In the quadratic benchmark $K'_+(0) = 0$, so every positive target blocking level induces a positive CIR best response. A shortcut fixed at $u = 0$ then omits a feasible partial reduction in effort.

Corollary 1 (Patience-threshold bias). *Fix $p^C > P$ and define the lower threshold generated by a candidate deviation value d as*

$$\delta(d; p^C, P) = \begin{cases} 0, & d \leq p^C, \\ (d - p^C)/(d - P), & d > p^C. \end{cases}$$

This function is nondecreasing in d and strictly increasing after the deviation becomes profitable. Therefore

$$\delta(d^{\text{exact}}; p^C, P) \geq \delta(d^0; p^C, P),$$

with strict inequality exactly when $d^{\text{exact}} > d^0$ and $d^{\text{exact}} > p^C$.

The combined threshold can remain unchanged when the CIR threshold moves because the CEN constraint may already bind. Section 5 identifies where the shortcut alters the binding constraint and where it reverses a sustainability classification. Primitive comparative statics must still recompute p_i^D , p_i^C , and P_i jointly; a change in deviation value alone does not determine the total change in δ_i .

5 Benchmark Implementability Results

The benchmark in Section 3.1 makes the individually-rational target set analytic before any grid is introduced. Let P_{CIR} and P_{CEN} be the Nash-reversion payoffs. Because $\pi(u) > 0$, the CIR strict-IR condition is

$$q[1 - a\pi(u)] - K(u) > P_{\text{CIR}} \iff a < a_{\text{CIR}}^{\text{IR}}(u) \equiv \frac{q - K(u) - P_{\text{CIR}}}{q\pi(u)}. \quad (19)$$

For the quadratic CEN cost, strict individual rationality requires

$$aD(u) - \frac{c}{2}a^2 > P_{\text{CEN}}.$$

Whenever $D(u)^2 > 2cP_{\text{CEN}}$, this is equivalent to

$$a \in (a_-^{\text{IR}}(u), a_+^{\text{IR}}(u)) \cap [0, 1], \quad a_{\pm}^{\text{IR}}(u) = \frac{D(u) \pm \sqrt{D(u)^2 - 2cP_{\text{CEN}}}}{c}. \quad (20)$$

The joint strict-IR set is therefore wedge-shaped: a must lie inside the censor's payoff interval but below the CIR boundary. Because both comparisons are strict, the set is open relative to the feasible action square; its closure is compact. Appendix E verifies these boundaries directly against payoff comparisons.

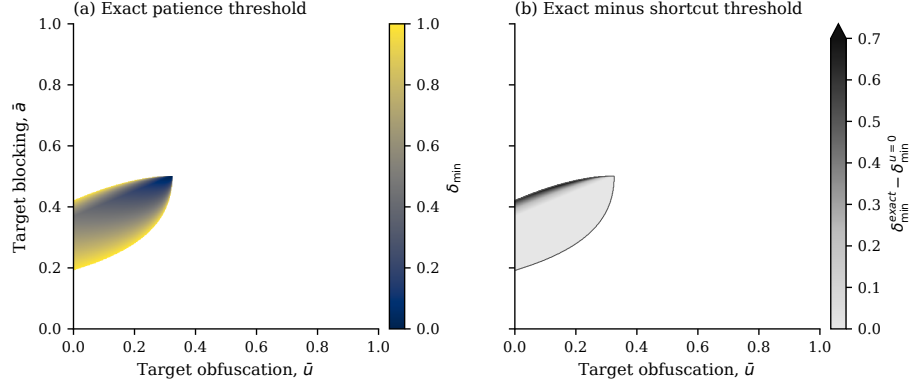


Fig. 2. Dynamic implementability in the illustrative benchmark. Panel (a) reports the exact combined patience threshold over the joint strict-IR region. Panel (b) reports the exact threshold minus the threshold obtained by fixing the CIR deviation at $u = 0$; values above 0.70 are saturated to preserve interior contrast. Blank profiles are outside joint strict individual rationality under Nash reversion.

We evaluate target geometry on uniform grids up to 2001×2001 . At the finest resolution, the joint strict-IR set occupies 6.46% of the normalized target square. Conditional on this set, the median exact combined threshold is 0.573 and its 90th percentile is 0.902. These statistics converge across grid sizes; Appendix Table A.2 reports the sequence. The percentages describe a uniform geometric measure over the normalized action square, not the probability that a real deployment is individually rational.

Figure 2 combines the two objects needed to read the dynamic result. Panel (a) maps the exact combined threshold. Panel (b) subtracts the threshold obtained when the CIR deviation is fixed at $u = 0$. The blank area lies outside joint strict individual rationality under Nash reversion, so the usual lower-patience interpretation does not apply there. Thresholds vary substantially inside the wedge because the deviation gain and continuation surplus change at different rates.

The shortcut changes the combined threshold on 18.62% of the strict-IR region. This share also marks targets at which the identity of the binding player changes: the exact CIR constraint binds, whereas the shortcut leaves the CEN constraint binding. At $\delta = 0.5$, the shortcut classifies 9.82% of the strict-IR region as sustainable even though the exact calculation rejects those targets. The bias is zero elsewhere in the combined threshold even though $u = 0$ remains a suboptimal CIR action; at those targets the unchanged CEN condition is tighter.

The representative target $(\bar{u}, \bar{a}) = (0.3, 0.5)$ shows why continuation surplus matters. The CIR's exact deviation retains positive obfuscation, $u^D \simeq 0.325$. The exact combined threshold is 0.711, while fixing the CIR deviation at $u = 0$ leaves the CEN constraint binding and gives 0.019. The CIR's one-period gain is only 0.000418, but its target payoff exceeds the Nash-reversion payoff by

only 0.000170. The high patience requirement therefore comes from a very thin continuation surplus, not a large absolute deviation gain.

This benchmark is descriptive of the model, not a calibration of a deployed CRS. Its design lesson is nonetheless sharp. Lowering the censor’s return $D(u)$ can expand the range of moderate blocking targets attractive to the censor, but maintaining a particular target also requires enough continuation surplus for both sides. An engineering design that looks favorable under a static classifier metric may remain dynamically fragile when either player is nearly indifferent between the target and credible reversion.

6 Monitoring, Granularity, and Design Implications

Noisy Public Monitoring. Suppose a public bad signal occurs with false-alarm probability ε under compliance and with probability $q_{\text{det}} > \varepsilon$ after any relevant deviation. False alarms lower the value of remaining in cooperation; missed detections weaken the link between deviation and punishment. For $G_i > 0$ and $L_i > 0$, the resulting threshold is

$$\delta_i^{\text{noise}} = \frac{G_i}{G_i(1 - \varepsilon) + (q_{\text{det}} - \varepsilon)L_i}. \quad (21)$$

A feasible threshold below one exists exactly when

$$(q_{\text{det}} - \varepsilon)L_i > \varepsilon G_i. \quad (22)$$

Perfect monitoring is the special case $(q_{\text{det}}, \varepsilon) = (1, 0)$, which recovers $G_i/(G_i + L_i)$. Appendix C gives the Bellman derivation. If detection probability depends on the chosen deviation, the action that maximizes current payoff need not maximize total deviation value; the incentive constraint must then be checked action by action. At the representative target, $(q_{\text{det}}, \varepsilon) = (0.6, 0.1)$ raises the combined threshold from 0.711 to 0.906, illustrating how a thin continuation surplus amplifies monitoring noise.

Discrete Implementation Modes. Continuous u and a are aggregate-intensity approximations. With coarse discrete modes, nearby deviations may disappear, so literal deviation values can differ from the continuous supremum. A finite deviation set must exclude the target action itself. Appendix D shows this distinction in the benchmark: on a 0.1 grid, $(0.3, 0.5)$ is a discrete stage equilibrium even though the continuous CIR has a nearby profitable deviation at $u \simeq 0.325$.

Security-Design Implications. First, a CRS should evaluate the censor’s payoff from blocking, not classifier accuracy alone. The strategic object $D(u) = BT_p\pi(u) - AT_b(u)$ shows that detectability and collateral exposure can both reduce blocking incentives. Second, technical feasibility must be paired with continuation surplus. A small $L_i = p_i^C - P_i$ can make an otherwise attractive target fragile even when the current deviation gain is small. Third, incentive tests

should match the implemented action space. Partial reductions can dominate disabling obfuscation when effort is continuous; genuinely discrete modes can remove those nearby deviations. Monitoring quality determines whether either design advantage can be enforced over time.

7 Conclusion

Obfuscation changes more than how accurately a censor classifies traffic. By moving effective detectability and the benign traffic exposed to blocking, it changes the censor’s net return $D(u)$ and therefore the equilibrium blocking response. The baseline game makes this link explicit while preserving boundary solutions and set-valued responses.

Repeated interaction adds a separate requirement. A desired obfuscation–blocking pair must give each side enough continuation surplus over a credible Nash-reversion path to offset its exact current deviation gain. Because a CIR can partially reduce a continuous, costly obfuscation action, testing only the action $u = 0$ can understate the binding patience requirement and reverse sustainability classifications. False alarms and missed detections tighten the same constraint by weakening the connection between compliance and continuation value. Designing a CRS against an adaptive censor therefore requires both a favorable blocking payoff and an enforceable operating regime.

References

1. Barradas, D., Santos, N., Rodrigues, L.: Deltashaper: Enabling unobservable censorship-resistant tcp tunneling over videoconferencing streams. *Proceedings on Privacy Enhancing Technologies* **2017**(4), 5–22 (2017). <https://doi.org/10.1515/popets-2017-0037>
2. Bocovich, C., Breault, A., Fifield, D., Serene, Wang, X.: Snowflake, a censorship circumvention system using temporary WebRTC proxies. In: 33rd USENIX Security Symposium. pp. 2635–2652 (2024)
3. Brückner, M., Scheffer, T.: Stackelberg games for adversarial prediction problems. In: *Proceedings of the 17th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. pp. 547–555 (2011). <https://doi.org/10.1145/2020408.2020495>
4. Dalvi, N.N., Domingos, P.M., Mausam, Sanghai, S.K., Verma, D.: Adversarial classification. In: *Proceedings of the Tenth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. pp. 99–108 (2004). <https://doi.org/10.1145/1014052.1014066>
5. Dritsoula, L., Loiseau, P., Musacchio, J.: A game-theoretic analysis of adversarial classification. *IEEE Transactions on Information Forensics and Security* **12**(12), 3094–3109 (2017). <https://doi.org/10.1109/TIFS.2017.2718494>
6. Dyer, K.P., Coull, S.E., Ristenpart, T., Shrimpton, T.: Protocol misidentification made easy with format-transforming encryption. In: *Proceedings of the 2013 ACM SIGSAC Conference on Computer & Communications Security*. pp. 61–72 (2013). <https://doi.org/10.1145/2508859.2516657>
7. Dyer, K.P., Coull, S.E., Shrimpton, T.: Marionette: A programmable network traffic obfuscation system. In: 24th USENIX Security Symposium. pp. 367–382 (2015)
8. Elahi, T., Doucette, J.A., Hosseini, H., Murdoch, S.J., Goldberg, I.: A framework for the game-theoretic analysis of censorship resistance. *Proceedings on Privacy Enhancing Technologies* **2016**(4), 83–101 (2016). <https://doi.org/10.1515/popets-2016-0030>
9. Fifield, D., Lan, C., Hynes, R., Wegmann, P., Paxson, V.: Blocking-resistant communication through domain fronting. *Proceedings on Privacy Enhancing Technologies* **2015**(2), 46–64 (2015). <https://doi.org/10.1515/popets-2015-0009>
10. Fudenberg, D., Maskin, E.: The folk theorem in repeated games with discounting or with incomplete information. *Econometrica* **54**(3), 533–554 (1986). <https://doi.org/10.2307/1911307>
11. Hardt, M., Megiddo, N., Papadimitriou, C.H., Wootters, M.: Strategic classification. In: *Proceedings of the 2016 ACM Conference on Innovations in Theoretical Computer Science*. pp. 111–122 (2016). <https://doi.org/10.1145/2840728.2840730>
12. Hu, Y., Chen, J., Zhu, Q.: Game-theoretic neyman-pearson detection to combat strategic evasion. *IEEE Transactions on Information Forensics and Security* **20**, 516–530 (2025). <https://doi.org/10.1109/TIFS.2024.3515834>
13. Khattak, S., Elahi, T., Simon, L., Swanson, C.M., Murdoch, S.J., Goldberg, I.: SoK: Making sense of censorship resistance systems. *Proceedings on Privacy Enhancing Technologies* **2016**(4), 37–61 (2016). <https://doi.org/10.1515/popets-2016-0028>
14. Maiti, A.: Block-a-mole: The sustainability frontier of moving-target censorship resistance (2026), preprint, arXiv:2606.08886 [cs.CR]

15. Musacchio, J.: A continuous strategy space adversarial classification game. In: Game Theory and AI for Security. Lecture Notes in Computer Science, vol. 16223, pp. 271–291 (2025). https://doi.org/10.1007/978-3-032-08064-6_14
16. Nasr, M., Farhang, S., Houmansadr, A., Grossklags, J.: Enemy at the gateways: Censorship-resilient proxy distribution using game theory. In: Network and Distributed System Security Symposium (2019). <https://doi.org/10.14722/ndss.2019.23496>
17. Nasr, M., Houmansadr, A.: GAME OF DECOYS: Optimal decoy routing through game theory. In: Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security. pp. 1727–1738 (2016). <https://doi.org/10.1145/2976749.2978367>
18. Tschantz, M.C., Afroz, S., Anonymous, Paxson, V.: SoK: Towards grounding censorship circumvention in empiricism. In: 2016 IEEE Symposium on Security and Privacy. pp. 914–933 (2016). <https://doi.org/10.1109/SP.2016.59>
19. Wails, R., Sullivan, G.A., Sherr, M., Jansen, R.: On precisely detecting censorship circumvention in real-world networks. In: Network and Distributed System Security Symposium (2024). <https://doi.org/10.14722/ndss.2024.23394>

A Baseline Best Responses and Static Proofs

Lemma 1 (Complete Best Responses). *Under Assumption S, $\mathcal{A}(d)$ and $\mathcal{U}(a)$ are nonempty compact intervals. An action a^* belongs to $\mathcal{A}(d)$ exactly when*

$$\begin{cases} d \leq C'_+(0), & a^* = 0, \\ C'_-(a^*) \leq d \leq C'_+(a^*), & a^* \in (0, 1), \\ d \geq C'_-(1), & a^* = 1, \end{cases}$$

and u^* belongs to $\mathcal{U}(a)$ exactly when

$$\begin{cases} qa\pi'_+(0) + K'_+(0) \geq 0, & u^* = 0, \\ qa\pi'_-(u^*) + K'_-(u^*) \leq 0 \leq qa\pi'_+(u^*) + K'_+(u^*), & u^* \in (0, 1), \\ qa\pi'_-(1) + K'_-(1) \leq 0, & u^* = 1. \end{cases}$$

The correspondence $\mathcal{A}(d)$ is nondecreasing in d in the strong set order, so its extremal selections after composition with $D(u)$ are nonincreasing in u . Extremal selections of $\mathcal{U}(a)$ are nondecreasing in a . Strict convexity of C makes $\mathcal{A}$ single-valued; strict convexity of K makes $\mathcal{U}$ single-valued uniformly.

Proof. For fixed d , maximizing $ad - C(a)$ is equivalent to minimizing the continuous convex function $C(a) - ad$ on $[0, 1]$. Convex optimality gives the three one-sided derivative conditions and a nonempty compact convex solution set. If $d_2 > d_1$ and $a_j \in \mathcal{A}(d_j)$, adding the two optimality inequalities yields $(a_2 - a_1)(d_2 - d_1) \geq 0$. Strict convexity gives uniqueness.

The CIR maximization is equivalent to minimizing $H_a(u) = qa\pi(u) + K(u)$, which is continuous and convex. The same one-sided argument gives the displayed conditions. For $J(u, a) = q[1 - a\pi(u)] - K(u)$, $u_2 \geq u_1$ and $a_2 \geq a_1$ imply

$$J(u_2, a_2) - J(u_1, a_2) - J(u_2, a_1) + J(u_1, a_1) = q(a_2 - a_1)[\pi(u_1) - \pi(u_2)] \geq 0.$$

Increasing differences order the extremal maximizers. Strict convexity of K makes H_a strictly convex for every a .

Proposition A.1 (Full static boundary statement). *Under Assumption S, the equilibrium correspondence has a fixed point. In addition:*

1. *The CEN response is uniquely zero for every u if and only if*

$$C(a) - C(0) > aD(0) \quad \text{for every } a \in (0, 1].$$

Either $D(0) < C'_+(0)$, or strict convexity of C together with $D(0) \leq C'_+(0)$, is sufficient. If K is uniquely minimized at zero, $(0, 0)$ is then the unique equilibrium.

2. The CEN response is uniquely one for every u if and only if

$$(1 - a)D(1) > C(1) - C(a) \quad \text{for every } a \in [0, 1].$$

Either $D(1) > C'_-(1)$, or strict convexity of C together with $D(1) \geq C'_-(1)$, is sufficient. Every equilibrium has full blocking exactly when no fixed point with $a < 1$ exists; uniform uniqueness of the CEN response is sufficient but not necessary.

3. If C and K are strictly convex, the equilibrium is unique.

Proof. By Lemma 1, both best-response correspondences are upper hemicontinuous with nonempty compact convex values. Kakutani's theorem supplies a fixed point of $\mathcal{U}(a) \times \mathcal{A}(D(u))$.

If $D(0) \leq C'_+(0)$, monotonicity gives $D(u) \leq C'_+(0)$ for all u . For $a > 0$,

$$[aD(u) - C(a)] - [-C(0)] \leq aD(0) - [C(a) - C(0)].$$

The first strict comparison in the proposition is therefore sufficient and, because it includes $u = 0$, necessary for uniform uniqueness. The stated derivative conditions imply it by convexity. At $a = 0$, the CIR chooses $\arg \min K$.

If $D(1) \geq C'_-(1)$, then $D(u) \geq C'_-(1)$ for all u . For $a < 1$,

$$[D(u) - C(1)] - [aD(u) - C(a)] \geq (1 - a)D(1) - [C(1) - C(a)].$$

The second strict comparison is sufficient and, at $u = 1$, necessary for uniform uniqueness. Strictness follows from the stated derivative alternatives. The exact “every equilibrium” condition is the absence of (u, a) with $a < 1$, $u \in \mathcal{U}(a)$, and $a \in \mathcal{A}(D(u))$.

Finally, with strict convexity write the single-valued responses as $A(u)$ and $U(a)$. If equilibria satisfied $u_2 > u_1$, then $a_2 = A(u_2) \leq A(u_1) = a_1$ and $u_2 = U(a_2) \leq U(a_1) = u_1$, a contradiction.

B Dynamic Incentives and Exact Deviations

Proof (Proof of Proposition 2). Compact actions and continuous payoffs imply bounded stage payoffs, so the one-shot deviation principle applies. Compliance has normalized value p_i^C ; a best current deviation followed by the stationary punishment has value $(1 - \delta)p_i^D + \delta P_i$. This gives Eq. (13). Rearranging under each payoff ordering gives the four stated cases. When $P_i > p_i^C = p_i^D$, continuity on an interval supplies distinct actions with current payoffs arbitrarily close to p_i^C ; for every $\delta > 0$, the continuation gain dominates a sufficiently small current loss.

Proposition A.2 (Scalar condition for an arbitrary current action). *Let a candidate action give current payoff x_i . The condition $p_i^C \geq (1 - \delta)x_i + \delta P_i$ has the following complete solution on $\delta \in [0, 1]$:*

1. If $P_i \leq p_i^C$ and $x_i \leq p_i^C$, it holds for every δ . If $x_i > p_i^C > P_i$, it holds exactly when $\delta \geq (x_i - p_i^C)/(x_i - P_i)$. If $x_i > p_i^C = P_i$, no $\delta < 1$ works.
2. If $P_i > p_i^C$ and $x_i \geq p_i^C$, no $\delta \in (0, 1)$ works. If $x_i < p_i^C < P_i$, it holds exactly when $\delta \leq (p_i^C - x_i)/(P_i - x_i)$.
3. If $x_i = P_i$, it holds exactly when $p_i^C \geq P_i$.

Proof. Write the inequality as $p^C \geq x + \delta(P - x)$ and solve the affine comparison under each ordering. The target-action maximum used in the main text rules out $x < p^C$ when $x = p_i^D$, leaving the four exact-deviation cases in Proposition 2.

Proof (Proof of Proposition 3 and Corollary 1). Because $u = 0$ is feasible, $d^{\text{exact}} \geq d^0$. Maximizing $\phi_{\bar{a}}$ is equivalent to minimizing the convex function $H_{\bar{a}}(u) = q\bar{a}\pi(u) + K(u)$. Zero minimizes it exactly when $H'_{\bar{a},+}(0) = q\bar{a}\pi'_+(0) + K'_+(0) \geq 0$, which is Eq. (18). A strict reversal makes a small positive u profitable.

For $d > p^C > P$, differentiation gives

$$\frac{\partial}{\partial d} \frac{d - p^C}{d - P} = \frac{p^C - P}{(d - P)^2} > 0.$$

Extending the threshold by zero for $d \leq p^C$ proves the corollary.

For completeness, the regular-region payoff derivatives are

$$\frac{\partial \delta_i}{\partial p_i^D} = \frac{p_i^C - P_i}{(p_i^D - P_i)^2} > 0, \quad \frac{\partial \delta_i}{\partial p_i^C} = -\frac{1}{p_i^D - P_i} < 0, \quad \frac{\partial \delta_i}{\partial P_i} = \frac{p_i^D - p_i^C}{(p_i^D - P_i)^2} > 0.$$

If a primitive θ moves all three payoffs, its total effect is

$$\frac{d\delta_i}{d\theta} = \frac{(p_i^C - P_i) dp_i^D/d\theta - (p_i^D - P_i) dp_i^C/d\theta + (p_i^D - p_i^C) dP_i/d\theta}{(p_i^D - P_i)^2}.$$

C Noisy Public Monitoring

Let the punishment state be absorbing with normalized value P_i . Under compliance, a bad public signal occurs with probability ε and moves play to punishment.

Proposition A.3 (Noisy-signal trigger). *The cooperation-state value is*

$$V_i^C = P_i + \frac{1 - \delta}{1 - \delta + \delta\varepsilon} (p_i^C - P_i). \quad (23)$$

If action x_i gives current payoff $U_i(x_i, \bar{x}_{-i})$ and bad-signal probability $q_{\text{det},i}(x_i)$, its incentive constraint is

$$(1 - \delta)[p_i^C - U_i(x_i, \bar{x}_{-i})] + \delta[q_{\text{det},i}(x_i) - \varepsilon](V_i^C - P_i) \geq 0. \quad (24)$$

With common $q_{\text{det}} > \varepsilon$ across relevant deviations and $G_i, L_i > 0$, Eqs. (21)–(22) are necessary and sufficient.

Table A.1. Discrete-action robustness

Grid (NEs)	Stage NE	Distance	Best deviations	Deviation payoffs	$\delta_{\min}$
0.25 (1)	(0.25, 0.50)	0.07555	(0.50, 0.75)	(0.602, 0.160)	0.000
0.10 (1)	(0.30, 0.50)	0.02556	(0.40, 0.60)	(0.618, 0.157)	0.000

Notes: Ordered pairs list (u, a) . Literal deviations exclude the target action. Each grid has one pure stage equilibrium; every equilibrium would be retained under multiplicity.

Proof. Compliance obeys

$$V_i^C = (1 - \delta)p_i^C + \delta[(1 - \varepsilon)V_i^C + \varepsilon P_i],$$

which yields Eq. (23). A unilateral action has value

$$(1 - \delta)U_i(x_i, \bar{x}_{-i}) + \delta[(1 - q_{\det,i}(x_i))V_i^C + q_{\det,i}(x_i)P_i].$$

Subtracting this value from compliance gives Eq. (24). Under common detection, substitute Eq. (23) and solve the binding constraint to obtain Eq. (21). Its denominator exceeds G_i exactly under Eq. (22). If detection varies by action, Eq. (24) must be checked for each action.

D Discrete-Action Robustness

Let $\mathcal{U}_h$ and $\mathcal{A}_h$ be finite action sets. A pure stage equilibrium is a mutual maximizer on these sets. A literal deviation excludes the target action, unlike the continuous convention in which the maximum including the target equals the supremum over arbitrarily close distinct actions. If the finite stage game has multiple equilibria, each candidate Nash-reversion payoff must be retained because punishment selection can change strict individual rationality.

In the benchmark, grids with spacings 0.25 and 0.10 each have one pure equilibrium. The finer grid selects $(u^N, a^N) = (0.3, 0.5)$, which is also the representative target. Its best literal CIR alternative is $u = 0.4$ with payoff 0.618356, below the target payoff 0.621632. The continuous action space instead contains the profitable deviation $u \simeq 0.3251$. This comparison shows why continuous intensity and discrete mode interpretations should not be mixed within one incentive calculation.

E Numerical Convergence and Regression Tests

The canonical command `python reproducibility/run_fc27.py` regenerates the equilibrium, all headline statistics, the main dynamic figure, and Table A.2. The calculation streams over target obfuscation values and evaluates grids from 151 to 2001 points per axis. It reports medians, quantiles, and shares rather than extrema near the open strict-IR boundary.

Table A.2. Grid convergence of stable benchmark statistics

N	Strict-IR (%)	Median δ	P90 δ	Changed (%)	False at 0.5 (%)
151	6.45	0.577	0.903	18.56	10.13
301	6.46	0.574	0.902	18.61	9.93
601	6.46	0.573	0.902	18.62	9.84
1001	6.46	0.573	0.902	18.62	9.83
2001	6.46	0.573	0.902	18.62	9.82

Notes: Each row uses an $N \times N$ uniform target grid. Shares are conditional on the joint strict-IR set except the Strict-IR column, which uses the full normalized target square. “Changed” records a change in the combined threshold; “False at 0.5” records shortcut-sustainable but exact-unsustainable targets.

The regression suite checks each closed-form response against numerical optimization. It also compares the fixed-point equilibrium with an independent root solve and checks KKT residuals. Further tests verify that the exact CIR deviation never yields less than $u = 0$, recover the perfect-monitoring limit, and compare the analytic strict-IR boundaries with direct payoff classification. The largest best-response discrepancy is 1.45×10^{-8} ; the equilibrium methods agree to 5.55×10^{-17} , and the largest benchmark KKT residual is 1.11×10^{-16} .